

Code-Layout, Readability and Reusability

Date	8 July 2026
Team ID	SWTID-2026-5447
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used 4 spaces for indentation throughout
2	Proper File Structure	Files and folders are logically organized	Yes	Separated frontend, backend, model folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used names like income, credit_score, age
4	Function / Method Names	Functions are descriptively named	Yes	Predict_credit_approval(), train_model()
5	Code Comments	Inline and block comments explain log c	Yes	Added comments for ML model and API routes
6	Modular Design	Code is split into reusable functions/modules	Yes	ML model, DB, and UI are in separate modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed duplicate functions and unused imports
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Used try-except for from validation and API

Reusable Components / Modules:

S No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Credit Prediction Model	Python, Scikit-learn	Can be used for loan Approval	High
2	Flask Web Application Module	Flask, HTML, CSS	Any ML prediction web app	High
3	Data Preprocessing Module	Python, Pandas	Any classification project	Medium
4	Form Validation Functions	JavaScript	All input forms	High
5	Model Saving/Loading Module	Python, Joblib/pkl	Any ML deployment project	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean folder structure with separate modules
Readability	4	Meaningful names and comments added
Reusability	4	ML model and web app modules are reusable
Documentation / Comments	4	Code documented with inline comments
Overall Score	4.5	Well structured and maintainable code